

Supplementary Methods and Results

Beyond the Score: Validating Evaluations of Synthetic IoT Time Series

This supplement provides the detailed methods and results referenced by the main study. It accompanies the thesis chapter on evaluating synthetic IoT time series. Cross-references to the main study use that study's internal numbering. Sections S1–S3 contain additional controls and full evaluation tables. Section S4 describes the research records, Sections S5–S7 give calculation details and implementation settings, and Section S8 reports reproduction checks and component results. Framework definitions, headline findings and their interpretation are given in the main study.

S1 Additional Metric and Control Checks

The following exploratory analyses extend the metric results in main study Section 7.1. They examine scaling, sample size, order changes and small value changes.

S1.1 Scaling Wasserstein Distance

The Wasserstein endpoint value of 1.464×10^3 came mainly from one sparse Zigbee event count whose TRAIN IQR was zero. The calculation used the 10^{-6} denominator safeguard, and that feature contributed 99.83% of the total at severity 1. The safeguard was used 15 times across three constructed dataset seeds and five severity levels. Excluding features with zero TRAIN IQR gave a mean endpoint response of 2.805. The median across all nine features, averaged over the three seeds, was 0.269. Restricting this median to features with positive TRAIN IQR gave 0.134. These summaries increased with severity. The metric meets the response rule, but its large scaled value is not a typical effect across features.

S1.2 Classifier Sample Size

For samples from the same distribution, AUC stayed near chance at 0.502–0.534 as sample size increased from 1,000 to 20,000 per class. Samples with controlled changes to individual value distributions remained almost perfectly separable. These comparisons assess classifier behaviour under known changes. They do not compare the residential generators assessed in the main study. The exploratory logistic regression diagnostics reported convergence warnings. Their summaries therefore describe the sample-size comparison without establishing fully converged logistic performance. These diagnostics are separate from the primary C2ST evaluation.

S1.3 Changes to Order and Values

The order check compares four controls, from unchanged data to changing block order, shuffling within blocks, and shuffling all rows. The ACF, frequency, and transition metrics gave ordered responses across these controls. The results concern the tested changes and do not establish that any one metric identifies temporal damage alone.

The value-change check alters values by 0.02 or 0.10 TRAIN IQR in the corresponding real controls. Protocol C2ST rose from 0.989 to 0.993 while median prediction loss relative to the real control remained

near one. Radio quality tasks were an exception, with median loss $4.30\times$ the control. A classifier can therefore detect small changes while most prediction tasks remain useful, and an average over tasks can hide a larger effect on one group.

These tests use the real-control loss ratio $A_{s,c}$ defined in main study Section 6.3.3.

S2 Residential Evaluation

S2.1 Target Availability and Task Selection

This section adds task-register details to the admission procedure in main study Section 6.1.1. Admission begins with the 199 task/model configurations selected during development. Contexts crossing joins in P0 block controls are excluded.

Validation retains 127 configurations: 42 same-time and 85 temporal, covering 17 groups. Of the 43 selected P0 representatives, 27 meet the checks. The search does not return to the 441 configurations excluded during development or replace failed P0 representatives. Before utility is calculated on TEST, its task set is fixed as the union of groups supported by the three matching alternatives. It contains 25 same-time configurations and nine temporal configurations. Each matching analysis reports only its own qualified groups. The nine temporal configurations do not support a primary temporal claim. The task register lists each member and the representatives supporting its group. Of the five admitted but ineligible groups, cross-source radio response and radio forecasting have no retained representative. Same-time physical context, packet-count bursts and physical-context forecasting have insufficient calibration or held-out support. The 25+9 scored configurations remain fixed. Primary qualification supports only the 25 same-time configurations.

S2.2 Matching and Threshold Sensitivity

Of the 32 profile variables used in main study Stage 2, thirty-one have observed TRAIN variation and contribute value and zero-rate descriptors. One contributes only missingness. Section S6 gives the rank and distance formulas.

Source indices 0, 2, 4, 6, and 8 calibrate the limits. Indices 1, 3, 5, 7, and 9 check them. These roles are identical across partner maps and do not depend on distances or prediction scores. The alternatives use the baseline partner map and the observed-rank partner map, with identical target handling. Table S1 reports decisions with and without the P0 bounds.

Table S1: Residential matching sensitivity. Counts are passing/eligible groups. All analyses use identical observed-target rules and fixed pair roles.

Partner map	Limits	Same-time	Temporal
Balanced observed, primary	Bounded	3/5	0/7
Balanced observed, primary	Unbounded	5/5	0/7
Baseline partners	Bounded	3/5	1/7
Baseline partners	Unbounded	5/5	1/7
Observed-rank partners	Bounded	3/5	1/7
Observed-rank partners	Unbounded	4/5	1/7

For the primary, baseline and observed-rank maps, mean matched distances on the same profile scale are 0.190, 0.225 and 0.214, respectively. The primary range is 0.064–0.245. Across 272 observed feature/pair comparisons per map, median KS distances are 0.069, 0.080 and 0.067. Each map includes a KS distance of 1. Missing comparisons are not treated as zero differences.

S2.3 Field Definitions and Generated-Value Checks

The value checks distinguish out-of-range values from values absent from the finite TRAIN sample but inside its range. A review of 71 flagged fields classified 37 as numerical measurements, durations, or rates and 13 as count fields. The other 21 remain unresolved. These classifications use available definitions and names, so field meanings and physical units are not fully verified. A numerical field does not fail solely because an in-range value was unseen. For a count field with few distinct values, integer validity is checked when the observed training values are integers. Fractional values in count fields with many distinct values are recorded for inspection. Unresolved fields retain conservative membership checks. The empirical range limits are not widened.

The residential generator uses binary sampling only when every finite value supplied to the sampling function is exactly 0 or 1. Inspecting all supplied values matters for sparse columns, whose initial records may all be zero even when later records contain other values. This is a rule for selecting the sampling operation, not a claim that the underlying physical variable is binary. Hurdle IID, independent stitching, shared stitching, and both learned overlays use this function through their base generators or rank-mapping operations.

The fifty runs from these five methods, across five seeds and two periods, also receive a range check over all 635 value columns. It compares generated values with the minimum and maximum of the exact prepared training input. The allowed violation fraction is 10^{-4} per column, and the observed maximum is zero. This check also requires finite output and binary observation masks. The 64-column learned-fit checks remain separate. All 90 source runs pass their applicable checks, but the 635-column range check covers only the specified fifty runs. Ten constructed software checks cover sparse nonbinary columns, order of input values, constant and binary inputs, the five methods using the function, and separation of range and membership errors.

The working-matrix composition is listed in Table S5. Section S8 reports the input and numerical-conversion checks. The task registry identifies targets, predictors, lags and learned/base membership. The 71-field review and value audit are also supplied. These numerical checks do not establish full semantic or physical validity.

S2.4 Prediction and Period Comparisons

The residential fitting and evaluation periods are specified in main study Table 3. Model-specific row caps are listed in Section S6. Table S2 extends main study Table 6 to all nine sources and both periods.

Before aggregation, completeness checks require unique task/period/seed/source keys and the expected configurations and five seeds in both periods. Each summary also requires a positive lower real-reference skill bound in the relevant period. Reproduction coverage is reported once in Section S8.

Table S2: Primary residential utility. Entries are loss ratios with 95% intervals conditional on the fixed tasks.

Source	Same period	Future period
Real row resampling	0.88 [0.84, 0.95]	1.10 [1.02, 1.21]
Real block control	0.78 [0.68, 0.87]	2.04 [1.94, 2.11]
Independent marginals	124.05 [120.61, 127.12]	95.17 [94.07, 96.28]
Column shuffle	123.77 [118.37, 129.65]	96.51 [93.80, 100.16]
Hurdle IID	122.13 [117.99, 126.54]	95.86 [94.83, 96.77]
Independent stitching	123.78 [115.54, 132.85]	96.42 [93.66, 98.84]
Shared stitching	16.25 [15.68, 16.94]	19.06 [18.60, 19.61]
GMM overlay	12.38 [10.78, 14.07]	14.12 [12.81, 15.73]
Latent VAR overlay	12.08 [10.74, 13.58]	11.81 [10.43, 13.42]

Decisions use unrounded interval endpoints. Section S8 adds direct baseline-relative skill, task-group contributions and raw-error components to these loss ratios.

Figure S1 supplies the period contrasts for all nine sources, complementing the selected results discussed in main study Section 7.3.

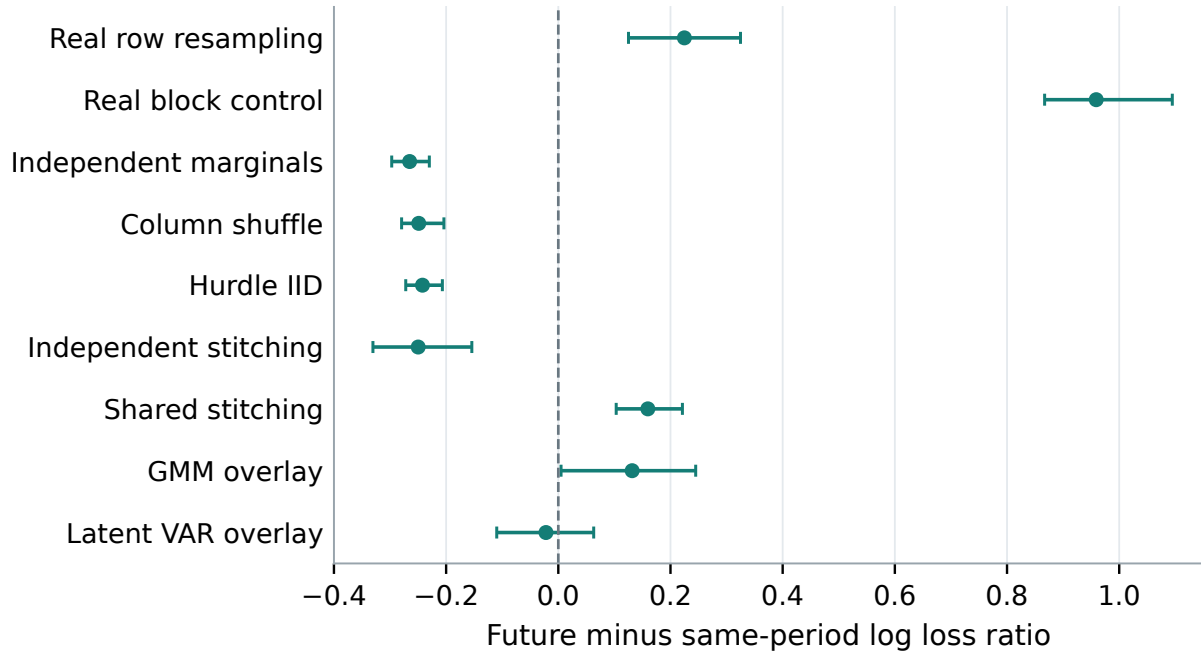


Figure S1: Residential change in log loss ratio from the same to the future period. Bars are 95% seed-bootstrap intervals for the three supported same-time groups. The dashed line marks zero change. Positive values mean greater relative loss later. Sources are refitted and each period has its own real reference. Negative values do not establish acceptable utility or a cause of the change.

S3 Detailed TON_IoT Checks

This section extends main study Section 7.4 with exact evaluation intervals, scope counts and full forecasting results.

S3.1 Task Admission and Periods

Table S3 gives zero-based, half-open intervals for every origin. Within primary TRAIN, admission fits on $[0, 13,776)$ and selects on $[13,801, 17,221)$. Each primary source generates 5,740 rows once for both evaluation periods.

At each additional origin, task admission uses only that origin's TRAIN period. The templates remain fixed.

Origins 0, 1 and 2 admit eight, eight and nine forecasting templates, respectively. The primary forecasting group passes with raw margin 1.023844 and applied margin 1.25. The primary burst task has only one usable calibration pair. Main study Section 7.4 reports the qualification outcomes and the unbounded origin-0 comparison.

Table S3: External periods. Intervals include the start and exclude the end. A 25-row gap separates adjacent fitting and evaluation intervals.

Split	TRAIN	Near evaluation	Future evaluation
Primary	[0, 17,221)	[17,246, 22,986)	[23,011, 28,752)
Origin 0	[0, 12,938)	[12,963, 17,275)	[17,300, 21,612)
Origin 1	[0, 14,376)	[14,401, 18,713)	[18,738, 23,050)
Origin 2	[0, 15,813)	[15,838, 20,150)	[20,175, 24,487)

S3.2 Metric and Source Checks

All eleven external controlled-response checks pass. The external generator functions differ from the residential functions as specified in Section S7.

S3.3 Grouped Classifier Resolution

Main study Section 6.3.2 defines the grouped C2ST procedure. Section S6 supplies its full model and randomisation settings.

The four scopes contain 17 IoT, 24 general-network, 23 IoT-labelled network and 48 balanced columns. Of the four primary summaries meeting the resolution conditions, two are anchor self-comparisons. Shared stitching has candidate-minus-anchor future AUC -0.00657 $[-0.01176, -0.00138]$. This conditional seed interval is not an equivalence test or calibrated directional significance test. All 72 summaries at each additional origin lack resolution. The primary outcomes are discussed in main study Section 7.4.

S3.4 Forecasting Utility

Table S4 gives the full future-period results for the seven sources passing output checks. Each has ten primary forecasting tasks in both periods and all five seeds. Six sources have an increased synthetic/real loss ratio in the future period. The paired change for the real block is inconclusive.

Table S4: TON_IoT future-period forecasting utility. Ratios use the matched real-trained reference. Intervals resample five seeds with the task set fixed. The loss margin is 1.10.

Source	Ratio	95% interval	Outcome
Real contiguous block	0.921	[0.861, 0.989]	Within margin
Independent stitching	1.330	[1.272, 1.393]	Higher loss
Shared stitching	1.396	[1.375, 1.415]	Higher loss
Real row resampling	2.003	[1.981, 2.030]	Higher loss
Hurdle IID	2.026	[2.014, 2.039]	Higher loss
Independent marginals	2.042	[2.035, 2.049]	Higher loss
Column shuffle	2.110	[2.000, 2.202]	Higher loss

Additional-origin utility uses one evaluation seed per origin and is not pooled into the primary confidence calculation.

S4 Research Records and Reproducibility

Saved records identify the data, method, split, task, scope, software, and seed for each calculation. File hashes check that reused records and numerical outputs agree with those identifiers. The accompanying material includes the aggregate qualification, source, utility, period-comparison, matching, and external

classifier tables used in the figures. The constructed benchmark has a separate 424-row summary of responses to increasing change.

The code and residential-data access statement is given in the associated study. Section S8 provides the comparison coverage and numerical tolerances.

S5 Calculation Details and Derivations

Main study Equations (1)–(8) define the workflow and primary comparisons. This section supplies implementation details and algebraic conditions needed to reproduce them. Numerical operating limits are collected in Section S6.

S5.1 Data Generation and Evaluation: Main Equations (1) and (2)

For implementation, the generator in main study Equation (1) can be written $G(X_d^{\text{train}}, \theta, s)$, where θ is the configuration and s the random seed. Configuration selection can use VAL-admitted targets, whereas parameter fitting and value transformations use TRAIN.

S5.2 Prediction Loss and the Baseline

For numerical targets, the implemented residential loss is

$$L = \frac{n^{-1} \sum_{i=1}^n |y_i - \hat{y}_i|}{\max\{Q_{0.75}(y) - Q_{0.25}(y), 10^{-6}\}}. \quad (\text{S1})$$

Here, y is the common evaluation target and Q denotes a sample quantile. The denominator is computed after fitting and cancels in common-target ratios and skills when loss floors are inactive. This is IQR-scaled MAE, not the training-difference MASE of Hyndman and Athanasopoulos [4]. A screen applied after scaling can still change task inclusion.

The binary counterpart is Brier loss [3], $n^{-1} \sum_i (y_i - \hat{p}_i)^2$. AUC and average precision are admission diagnostics and do not replace this loss in the ratios. Main study Section 6.1.2 specifies both constant baselines.

S5.3 Matched-Block Calibration: Main Equation (3)

Let R_A and R_B be ratios from two independently sampled real block controls. Both controls are drawn from the source member of a matched pair and evaluated on its target member. They are not the two members of that pair. Their discrepancy is

$$|\ell_A - \ell_B| = |\log R_A - \log R_B| = |\log(R_A/R_B)|. \quad (\text{S2})$$

Thus, doubling and halving give the same absolute discrepancy, $\log 2$. Using logs is appropriate for a multiplicative comparison. It also avoids treating the reciprocal changes 2 and 1/2 as unrelated additive amounts.

The nested quantiles in main study Stage 2 use NumPy linear interpolation. Taking the 90th percentile over seeds within each pair and then over pair summaries is not equivalent to pooling all 25 values.

For temporal calibration, $(\ell_A + \ell_B)/2 = \log \sqrt{R_A R_B}$. Exponentiating the nested quantile returns a bound on the loss-ratio scale before applying the limits in Section S6.

S5.4 Real-Reference Skill: Main Equation (4)

For main study Equation (4), the positive denominator preserves the sign of $L^{\text{naive}} - L^{\text{real}}$, including when the 10^{-12} floor is active. Interpretation as a fractional reduction requires baseline loss above the floor. The same condition applies to synthetic skill in Equation (7).

S5.5 Period Comparison, Synthetic Skill, and Loss Ratios: Main Equations (5)–(8)

Before forming main study Equation (6), the residential code requires positive baseline loss and excludes reference loss below 10^{-5} . This screen applies to IQR-scaled regression loss or Brier loss. None of the 3,060 final residential comparison records is excluded by it. A zero synthetic loss is flagged, and logged ratios use $\log(\max(R, 10^{-12}))$.

Equation (5) compares this ratio across periods:

$$\Delta = \log R^{\text{future}} - \log R^{\text{same}} = \log(R^{\text{future}}/R^{\text{same}}). \quad (\text{S3})$$

Thus $\exp(\Delta)$ is the factor of change in relative loss. Section S8 decomposes it into real- and synthetic-error components before aggregation.

For Equation (8), let a denote the selected real-data control, evaluated with the same seed s and configuration c as the candidate. With the same TRTR denominator L_r , task, seed, and test targets, let L_{syn} and L_a denote the candidate and control losses. Then

$$\frac{R_{\text{syn}}}{R_a} = \frac{L_{\text{syn}}/L_r}{L_a/L_r} = \frac{L_{\text{syn}}}{L_a}. \quad (\text{S4})$$

The cancellation is valid only for matched denominators. If the real references differ, then

$$R_{\text{syn}}/R_a = (L_{\text{syn}}/L_a)(L_{r,a}/L_{r,\text{syn}}).$$

Reference identity must therefore be checked explicitly. Equal within-pair counts alone do not justify cancellation. This additional comparison separates the candidate's loss from loss already present in the real control. It does not replace the primary TRTR comparison or Stage 3A.

S5.6 Other Summaries

A geometric mean is $\exp(n^{-1} \sum_i \log R_i)$. It combines multiplicative changes and treats reciprocal ratios symmetrically. Within each seed, the residential calculation averages log ratios across configurations in a task family, then averages families within each group and gives groups equal weight. The external calculation uses the median log ratio within each family, then averages families within each group. Both calculations balance families, but they differ in how they combine configurations within a family. Repeatedly sampling seeds gives intervals conditional on these fixed task sets.

C2ST excess subtracts the real-control AUC from the candidate AUC using the classifier two-sample approach [5]. Its interpretation and resolution requirements are given in main study Section 6.3.2.

S6 Basis of Numerical Settings and Model Choices

Recorded counts, data-derived settings, and fixed choices must be read differently. Counts describe the data or completed design. A data-derived setting follows a stated calculation on TRAIN or selection on VAL. A fixed setting is a tolerance, model size, or computing budget. The reasons below explain each setting's role and tradeoff. Where no selection study is retained, they do not establish that the exact value was optimal or that this explanation was documented before testing. Method references support the method, not the study's particular cutoffs.

S6.1 Data, Splits, and Matching

Table S5: Data and sampling choices.

Setting	Basis and limit
Residential working matrix	635 evaluation values, 356 associated masks and 19 additional task/scope columns give 1,010 columns. This differs from the 439-mask registry.
One-second residential and five-second TON_IoT grids	The residential table uses its recorded one-second grid. The external five-second grid combines sources with different reporting rates. It is a fixed compromise between detail and sparse observations.
Chronological periods	Time-series evaluation guidance [2, 4] motivates temporal separation. Main study Table 3 gives the primary periods. Table S3 gives all external origins. These are fixed allocations, not optimised proportions.
121 and 25 excluded rows	Derived from maximum lag plus maximum horizon plus one current row: 60 + 60 + 1 at one second, and 12 + 12 + 1 at five seconds. They prevent task-window overlap. They do not prove independence between adjacent blocks.
Three additional origins	TON_IoT repeats admission, P0 qualification, and evaluation at three origins with earlier training data. Their row intervals are in Section S3. Each uses one evaluation seed and is descriptive. Residential outputs have not been evaluated at these additional origins.
60-second capture-session gap	A fixed session rule equal to twelve five-second intervals. The longest common recorded-time session is selected, and no gap is closed by concatenating separated periods. This is distinct from train/test embargoes and is not proof of continuous acquisition.
20 residential blocks of six hours	Gives ten non-overlapping source/target pairs. Six hours greatly exceeds task windows while leaving enough blocks in TRAIN.
32 profile variables	Selected from TRAIN to describe the blocks. Bounded ranks, observed zero fractions, and missing fractions make the profile independent of raw measurement units. Constant observed TRAIN fields contribute only missingness. The count controls profile size and does not state statistical degrees of freedom.
Matching roles	Source indices 0, 2, 4, 6 and 8 calibrate, while 1, 3, 5, 7 and 9 qualify. Section S2 uses these same roles for every partner map. No maximum-distance equivalence rule is imposed.
TON_IoT 30-minute blocks and 95% availability	Each origin uses its own TRAIN-only matching. Candidate durations are 30, 60, 90, 120, 150 and 180 minutes. The first supporting ten disjoint screened pairs is selected. All four select 30 minutes in this run. The 0.95 finite-value screen is an explicit proxy, not independently measured acquisition coverage.

For each varying profile field, the rank of an observed value x is $r(x) = (n_{<x} + n_{\leq x})/(2n)$, where n counts finite observed TRAIN values. This is the midpoint of its empirical percentile interval and lies in $[0, 1]$. Block descriptors average these ranks and separately record observed zero rates and missing fractions. A block without observations uses 0.5 for value and zero-rate descriptors and records missing fraction 1. Time-of-day sine and cosine means are shifted and scaled to $[0, 1]$. This choice avoids comparing raw units, but midpoint replacement is a declared matching convention rather than a measured value.

Let z_{ik} denote descriptor k in block i , with G nonempty descriptor groups and n_g descriptors in group g . The distance is $d(i, j) = \sqrt{\sum_k (z_{ik} - z_{jk})^2 / (G n_{g(k)})}$. This is a weighted Euclidean distance with equal total weight per group. One-to-one assignment minimises the sum over ten selected pairs. The formula expresses the chosen equal-weight rule. It does not derive a universal match-quality threshold. Section S2 gives the observed distances and residual distribution differences.

S6.2 Metric Response and Acceptance Rules

Table S6: Decision limits, their purpose, and available checks.

Setting	Basis and available evidence
Three metric-test seeds, 24,000 rows, and five severity levels	Seeds 19, 29, 39 and levels 0, 0.25, 0.50, 0.75, 1 give repeats and an ordered response curve. Rows support many temporal windows while bounding runtime. These are constructed checks, not a sample-size calculation for real-world error rates.
Spearman $\rho \geq 0.80$	Requires a strongly ordered increase across severity. The smallest recorded target response was 0.8944. This supports that particular boundary, not every possible metric or perturbation.
Endpoint minima	The distributional minima are KS 0.02, Wasserstein/IQR 0.05, zero-rate error 0.002, and C2ST excess 0.02. The temporal minima are ACF error 0.01, spectral error 0.03, and transition error 0.0005. The between-column minima are correlation error 0.02 and co-activation error 0.0005. These fixed minima exclude tiny ordered changes on different metric scales.
Same-distribution C2ST excess ≤ 0.03	A modest tolerance for the diagnostic control rather than an exact type-I-error level. Raw AUC was 0.506 with recorded mean excess zero. Later sample-size controls gave 0.502–0.534.
C2ST ceiling 0.995	Marks almost complete separation, with 0.9945 as a sensitivity alternative. Main study Section 7.4 reports the resulting resolution outcomes.
90th percentile, taken twice	An upper summary of control variation with equal pair weight, as derived in Section S5. Five seed values and five pair values make it coarse. No nominal 90% error guarantee is claimed.
Same-time margin: $\log(1.10)$ to $\log(1.50)$	The floor prevents an almost zero tolerance. The cap prevents very poor control transfer allowing an arbitrarily large difference. Values correspond to multiplicative limits of 10% and 50%.
Temporal bound: 1.25 to 1.80	Allows a wider tolerance for sampled-block transfer and boundary effects while retaining a cap. Table S1 reports unbounded sensitivity.
At least four of five calibration pairs and four of five check pairs	Permits one unsupported pair in each set while requiring several blocks. Four is a minimum coverage screen, not evidence of accurate asymptotic inference.
Stage 3A lower 95% skill endpoint > 0	Zero is the meaningful boundary between improvement and no improvement over the simple baseline. Requiring the lower endpoint above it accounts for observed pair/seed variation. Removing this check accepts four KT10 cases labelled for withholding.
Identity interval within $[-0.03, 0.03]$	About 3% multiplicative tolerance in either direction for matching and fitting variation. It is a statistical control for repeated fits. Exact cached-prediction checks require numerical equality up to arithmetic tolerance. The 0.03 policy must not substitute for them. No separate sweep is recorded.
Balanced mean within-margin rate at least 0.75	The code averages the within-margin indicators within task families, then across families, the two block controls, matched pairs, and seeds. This is a weighted repeat average, not a percentage of distinct configurations. Three quarters is a fixed requirement for these repeated control comparisons.
Order-loss lower endpoint $\geq \log(1.05)$	A fixed minimum of 5% extra loss after removing order. The precise boundary comparison is given below.
At least 70% of variable columns remain variable	A coarse screen against widespread collapse in the learned scope. The residential fits pass this screen. It does not ensure good utility or prove that every individual column retains variation.

Table S6 continued

Setting	Basis and available evidence
Support violations $\leq 10^{-4}$	A fixed per-feature tolerance of one entry per 10,000. Range violations are counted separately from unseen in-range values under the field rules. The 635-column range check finds zero violations in the fifty residential runs specified in Section S2. It is not a verification of physical ranges.
Activity errors: mean 0.08, 95th percentile 0.20, maximum 0.40	Check overall error, the high-error tail, and the worst feature in percentage-point terms. Residential learned scopes pass. TON_IoT learned sources fail their applicable output checks. The same declared numerical limits apply to these decisions.
Utility upper 95% ratio endpoint ≤ 1.10	A stated allowance of at most 10% additional loss. It defines the outcome called within the loss margin, rather than generator validity. Shared stitching’s residential future-loss lower endpoint 18.60 is far above this margin.

For protocol-other forecasting, the unrounded lower endpoint of the row-resampling minus mean block-control log ratio is 0.037919, equivalent to approximately 3.865% additional loss, compared with $\log(1.05) = 0.048790$. A zero threshold would change this predicate. It would not change the matching, fitting or primary policy.

S6.3 Constructed Data and Controlled Changes

The metric-test data contain dependent continuous values, periodic activity, counts, binary states, and sparse events. A stable autoregression with coefficient 0.96 and Gaussian innovation scale 0.25 creates persistence. A sine component with period 3,600 rows creates a repeated pattern. Conditional Poisson, binomial, and lognormal draws connect traffic and activity. These are planted patterns for checking metric responses, not parameters estimated to represent an actual household. We use these fixed constants to create repeatable tests with known patterns. They do not describe a third observational dataset or claim to reproduce a particular home.

At severity u , selected column values are changed by $x' = (1 + 1.5u)x + u \text{IQR}(x)$, and selected sparse columns have a fraction $0.60u$ of nonzero entries set to zero. The coefficients make a clear graded change in scale, location, and sparsity. No claim depends on them being the only suitable coefficients. For partial order damage, a fraction u of rows is replaced using a common randomly drawn row index across columns. At full damage, we use a permutation. For column alignment damage, the replacement indices differ by column. Partial replacement can also change value frequencies, and column changes can affect time patterns. The six observed cross-responses therefore limit claims of specificity.

S6.4 Task Definitions and Prediction Models

Table S7: Task and predictor settings.

Setting	Role and limit
Residential lags of 1, 5, 30, and 60 seconds and horizons of 5, 15, and 60 seconds	Cover immediate, short, and longer delays within a minute. They are fixed task grids, not estimated device response times. Event windows use 30 or 60 seconds, and cross-source windows use 30 seconds.
Burst threshold: TRAIN 99th percentile	Defines unusually high values relative to training behaviour. The percentile is computed from TRAIN, while the choice of the 99th percentile is a fixed rare-event definition. It is not a labelled attack boundary.

Table S7 continued

Setting	Role and limit
Task construction budgets	Caps of two targets per source/signal group and 32 predictors, or 24 for cross-source tasks, limit computation and redundancy. Admission retains 127 of the 199 residential configurations selected during development and 27 of 43 P0 representatives, without replacing failed representatives. TON_IoT screens 32 previously defined templates and retains eleven on the primary split, of which ten enter the qualified forecasting group. These are selection counts, not independent replications.
At least 300 usable training and 300 usable validation examples	Rejects very small tasks before repeated fitting. Even 300 rows can provide few rare events, so further class-support and pair checks are required. This is not a claim of statistical sufficiency.
Positive prevalence 0.005–0.50	Excludes extremely rare or majority-positive event definitions in the validation screen. At the lower edge, few positive examples may remain. Interval precision is not guaranteed.
At least 10% improvement, AUC 0.60, and average precision gain 0.01	Loss improvement screens usefulness relative to a constant baseline. AUC checks ranking above chance. Average precision gain above the validation event rate checks rare-event ranking. The three tests answer different questions. Their exact cutoffs are fixed screens, not tuned acceptance probabilities.
Residential prediction forest	80 trees, maximum depth 16, minimum leaf size 2. A random forest [1] captures nonlinear interactions. Tree count, depth, and leaf size bound fitting cost and complexity.
Residential ridge regression	Standardised inputs and penalty $\alpha = 1$. The iterative least-squares solver (LSQR) uses tolerance 10^{-6} and a maximum of 10,000 iterations. Regularisation stabilises fitting with related predictors. Standardisation makes the penalty comparable across input scales. The unit penalty and solver limits are fixed settings.
Residential logistic regression	Standardised inputs and default penalty strength $C = 1$. The limited-memory quasi-Newton solver (L-BFGS) allows at most 600 iterations. Logistic regression models binary-event probabilities through a linear predictor. The limit bounds fitting time, but convergence warnings must still be reported.
TON_IoT task models	The evaluation uses the residential ridge and logistic model definitions, including standardisation fitted only on training inputs. Missing predictors retain the prepared zero-filled values. This differs from the residential median imputation with missingness indicators.
Task row caps	Up to 60,000 fitting rows and 30,000 scoring rows, with a separate metric cap of 50,000. These bound repeated computation. They do not establish equal effective sample sizes under dependence.

S6.5 Why These Generators and Configurations?

Main study Section 6.2 explains the generator selection. The settings below specify the residential implementations, including the latent VAR model [6]. Section S7 gives the external differences.

Table S8: Residential generator settings. External differences are in Section S7.

Setting	Role and limit
64 learned columns	Prioritise VAL-admitted targets, then fill by TRAIN variability across source/signal groups. Within each group, the score is TRAIN standard deviation times the square root of the nonzero fraction, favouring variation with activity. This is a chosen ranking, not fitted feature importance. The selected counts are 20 telemetry, 18 wireless, 15 router, and 11 Zigbee columns. Sixty-four bounds model size. The learned models do not cover every value column.
100,000 fitting rows and 70,000 generated rows per period	Fixed computation caps. The output exceeds the 60,000-row task training cap plus the 121-row window. GMM subsamples training rows. VAR uses a consecutive training window so lags remain meaningful.
Real block and stitch lengths	For requested output n , the default segment length is based on $\max(4 \times 121, n/30)$, with integer rounding and bounds in the saved function. At $n = 70,000$ it is 2,333 rows. P0 instead uses $\max(4 \times 121, \min(n_{\text{anchor}}/5, 3600))$, giving 3,600-row sampled segments within a 21,600-row source block. These are different lengths for different purposes. The multipliers and caps are fixed construction settings, not estimated optimal block sizes.
Signed log transform and standardisation	$\text{sign}(x) \log(1 + x)$ reduces the influence of large counts while retaining sign. TRAIN means and standard deviations then balance scales. Inverse transformations return to original units.
Sixteen principal component analysis (PCA) components	Reduces 64 columns before fitting. The fixed rank controls model size and overlap in the information carried by columns. Both models use randomised PCA with the run seed.
Eight GMM components and diagonal covariance	Represents several row regimes with fewer covariance parameters than a full-covariance mixture. Eight is a fixed size. Each component remains multivariate through the common PCA representation.
GMM covariance floor 10^{-5} , two starts, and maximum 250 iterations	The floor avoids singular covariance estimates. Two starts reduce reliance on one initial fit and the iteration limit bounds work. The saved code rejects non-convergence. Convergence alone does not establish an optimal fit.
VAR lags 1, 5, 30 and ridge penalty 1	Models several short delays without a long autoregression. These are seconds in residential data but 5, 25, and 150 seconds in TON_IoT. Exact lags and penalty are fixed, not selected by information criteria.
Residential VAR fitting and prediction check	Within its 100,000-row fitting window, the first 80,000 rows fit PCA and VAR and the remaining 20,000 check prediction. This internal allocation reserves later observations for checking the fitted model. It is separate from task admission and downstream evaluation. PCA must retain at least 55% of transformed variance. Validation MAE must be at most 1.25 times the better of persistence and latent-mean baselines. These fixed fit screens differ from the downstream 1.10 utility rule. The external VAR does not use this internal split.
VAR numerical settings	LSQR ridge uses tolerance 10^{-6} and maximum 10,000 iterations. Simulation adds sampled TRAIN residual vectors and clips latent states to TRAIN 0.5th–99.5th percentiles to limit unstable excursions. These stabilisers are part of the generator, not evidence that all temporal structure is preserved.
Rank mapping and bases	Generated ranks determine the order of empirical hurdle draws, assigning values within the allowed training support. GMM uses its fitting sample for this mapping and a hurdle IID base. VAR uses full TRAIN and a shared-stitching base. A final projection enforces support constraints. These different mappings and bases are parts of the evaluated pipelines.

S6.6 Classifier Settings, Repeats, and Benchmark Size

The external C2ST uses a 100-tree forest with maximum depth 12, minimum leaf size 3, and balanced weights, plus standardised logistic regression. Three group-disjoint splits use 35% for evaluation. Five-minute groups contain 60 rows at the external five-second cadence. Groups are paired across sources before sampling and boundary context is trimmed. The cap is 8,000 rows per class. These fixed settings limit repeated computation and allow both linear and nonlinear separation. They do not establish independence between groups or a universal test level.

The smaller randomisation forest has 25 trees and nineteen label randomisations. It is a diagnostic, not the null distribution of the maximum statistic from the two main classifiers. Residential metric controls use 300-row groups and a 25,000-per-class cap, with the separate 1,000–20,000-per-class sensitivity check described in Section S1. These controls are distinct from a classifier comparison of the evaluated residential generators.

The main seeds 1337–1341 fix algorithmic random choices. Bootstrap draws resample existing pairs and seeds. They do not generate new deployments. The 95% percentile interval convention uses the 2.5th and 97.5th percentiles of 5,000 draws. Crossed resampling [7] reflects two repeated dimensions, but five pairs and five seeds give limited information, and no population or simultaneous-coverage guarantee is claimed.

The 18 benchmark scenarios share four base-sequence seeds, giving 72 cases. Five evaluation seeds and two prediction models provide repeats within each case. Main study Sections 6.4.3 and 9 explain the development role of these scenarios.

Each base sequence in the 18-scenario benchmark has 32 value fields on a six-second grid. It contains 72,000 TRAIN, 24,000 VAL, and 24,000 TEST rows, plus two 121-row gaps, giving 120,242 rows. Twenty 3,600-row TRAIN blocks each span six hours and provide ten matched pairs, divided five for calibration and five for checking. These counts support the chosen repeated-block design. The 32-field width, planted autoregressive coefficients, and perturbation sizes are construction settings, not household estimates or values selected as optimal.

The benchmark recomputes the metric checks for working metrics. For cases representing flat and reversed metric responses, we directly insert a failed-check flag. Those cases test whether a failed metric check prevents acceptance. They do not test whether the numerical response check detects a faulty metric. Likewise, the learned-output failure cases collapse columns or insert zeros into real-derived records before applying the output checks. They do not fit a learned generator.

Performance-outcome coverage comprises 16 within-margin utility cases, four higher-loss utility cases and four unresolved closeness cases. The remaining 48 cases have no applicable performance outcome. Utility covers the future-period arm only. The KT01–KT18 register records the constructions and inserted faults.

Ablations reuse saved evidence without refitting. The alternative outcome rule uses median synthetic-model skill while retaining P0 and Stage 3A. Because all final labels agree, bootstrap intervals for their agreement are degenerate. Main study Sections 7.3.1 and 8.2 discuss the interpretation.

S7 Implementation Differences

The residential and external studies ask the same evaluation questions but differ in data preparation and generators. Table S9 lists these differences and the comparisons they limit.

Table S9: Implementation differences between residential and external evaluations.

Item	Recorded implementation and implication
Missing predictors	Residential models use fitted medians and explicit missingness indicators. External models use zero-filled predictors. Unobserved real targets are excluded in both. Generated observation masks are tested only as binary outputs in the residential study, without controlled missingness qualification.
P0 and block controls	Residential controls join real segments and exclude task contexts crossing joins. External P0 controls are contiguous. Nested 90th percentiles supply limits, and caps remain decision conventions.
Learned-column selection	Residential uses a 64-column selection based on VAL-admitted targets and TRAIN variability. External models use 48 fixed columns. These are partial learned models, not complete modern-generator benchmarks.
Binary sampling	Residential binary detection checks every finite value supplied to the function used by all three hurdle methods and both learned overlays. The external generator functions are separate and do not use this function.
Stitching	Residential stitching orders hurdle draws by real-segment ranks. External stitching copies 60-row segments, or five minutes. We refer to these as residential rank stitching and external segment-copy stitching. They are different implementations.
External clipped GMM	Standardises 48 columns and fits eight diagonal mixture components without residential PCA. It uses at most 150 iterations, covariance floor 10^{-5} , and one initial fit. Outputs are clipped to observed ranges, without empirical rank mapping.
External clipped latent VAR	Uses standardisation, up to sixteen PCA components, lags 1, 5, and 30, and ridge penalty 1. Simulation adds Gaussian noise with standard deviation 0.05, not sampled residual vectors. Final values are clipped to TRAIN ranges. The noise amplitude remains an untested fixed setting.
Output scopes	Residential learned-fit checks cover 64 learned columns. The 635-column range check covers the fifty runs of the three hurdle methods and two learned overlays. External learned and whole-output diagnostics are recorded separately. Neither proves all-field physical or semantic validity.
Utility aggregation	Residential summaries average log ratios within task families. External summaries use within-family medians. Both then give families equal weight within a task group. This difference and the distinct task sets prevent a direct numerical ranking across datasets.

S8 Numerical Verification and Interpretation

The following checks extend the reproduction summary in main study Section 6.4.2. Table S10 collects the coverage counts. The text specifies comparison rules and additional component results.

S8.1 Fresh Runs and Result Comparisons

The TON_IoT comparisons cover admission, qualification, source checks, classifier summaries and utility across the primary split and three additional origins. Prediction-loss checks include diagnostic losses for failed outputs. No source, task, admission or classifier execution error is recorded. Recalculation of the constructed benchmark from 8,000 saved real-comparison rows reproduces twenty Stage 3A group estimates and interval endpoints.

Residential checks additionally reproduce 90 utility summaries and require unique records with identical comparison keys. Numerical comparisons use absolute tolerance 10^{-7} and relative tolerance

10^{-6} , while categorical outcomes must agree. The largest absolute difference in task-level loss ratio is 2.15×10^{-10} . Prediction losses are recalculated from targets, predictions and baseline training targets. Task discovery remains fixed to the declared candidate set, within which admission and evaluation are repeated.

Table S10: Completed author-side reproduction checks. Counts refer to computational comparisons, not independent datasets.

Workflow	Comparison coverage and outcome
Saved calculations	144 checks pass, including 45 printed headline values.
Constructed experiment	Full scenario and metric rerun passes its result comparisons.
TON_IoT evaluation	All 72 source jobs complete, and 87 comparison checks pass.
TON_IoT predictions	8,508 loss comparisons pass.
Residential evaluation	P0 is refitted, and 90 source runs, 3,060 utility records, and 132 qualification records reproduce.
Residential predictions	16,291 distinct prediction-file loss checks pass.

S8.2 Input and Output Consistency

The residential input matches the recorded file and contains 740 stored columns, including the 664 raw fields required for the 1,010-column working matrix. The 439 registered missingness counts reproduce, with 114,941,541 nonfinite cells. All 1,992 field-by-split diagnostics agree, with no float32 overflow or nonzero-to-zero underflow. The 136 real-training/evaluation context counts cover 34 task/model configurations, two periods and two roles. The full-range check specified in Section S2 gives 31,750 field/run comparisons with zero violations.

The external prepared table has a regular five-second grid. Its masks identify observations before filling, and modelling values follow the stated zero-fill rule. Prepared-table comparisons check the schema, row order, timestamps, missingness, counts, and values. They allow a one-step floating-point difference only for the packet-length standard-deviation fields. Other parsed fields must match. This accommodates the verified numerical difference between platforms without changing model-result tolerances. Compressed-file hashes are recorded separately from these content comparisons.

Saved TON_IoT records also permit checks of training and evaluation targets, source indices, fitted scaling parameters, and linear-model predictions. Recalculation reproduces the 14 primary utility summaries and confirms the 16 learned-output failures. These failures remain scientific outcomes of the output checks even though their computations complete successfully. Fresh prediction losses are checked against their saved targets and reported references. Byte identity of prediction arrays across executions is not required.

S8.3 Direct Skill and Period Components

For one matched task and seed, let B be the loss of the real-training constant baseline and let E_r, E_s be the real- and synthetic-trained losses on common observed targets. For positive B, E_r and inactive floors,

$$R = E_s/E_r, \quad S_r = 1 - E_r/B, \quad S_s = 1 - E_s/B, \quad S_s = 1 - R(1 - S_r).$$

We compute skills before aggregation. Residential skills follow the same equal-family, equal-group hierarchy as utility, but average skills rather than log ratios. The five-seed bootstrap keeps the tasks and weights fixed and uses 5,000 resamples with the recorded seed 901. It describes repeated computation on this selected collection, not a population of homes.

Table S11: Direct skill against the training-median baseline on the 25 primary residential same-time configurations. Brackets are conditional 95% seed-bootstrap intervals.

Source	Same-period skill	Future-period skill
Real row resampling	0.908 [0.907, 0.909]	0.922 [0.913, 0.928]
Real block	0.857 [0.813, 0.892]	0.645 [0.607, 0.674]
Independent marginals	-0.620 [-0.927, -0.380]	-0.222 [-0.237, -0.207]
Column shuffle	-0.681 [-1.036, -0.367]	-0.258 [-0.341, -0.213]
Hurdle IID	-0.559 [-0.721, -0.422]	-0.233 [-0.241, -0.225]
Independent stitching	-0.864 [-1.513, -0.345]	-0.294 [-0.330, -0.258]
Shared stitching	0.476 [0.366, 0.584]	0.523 [0.439, 0.606]
GMM overlay	0.662 [0.605, 0.713]	0.545 [0.496, 0.594]
Latent VAR overlay	0.637 [0.570, 0.699]	0.599 [0.483, 0.682]

The combined real-reference skills are 0.9004 and 0.9281 in the same and future periods. Table S11 gives every source’s direct skill, and Table S12 separates the group contributions to future loss ratios. Main study Section 8.2 explains why these comparisons can give different impressions of performance.

Table S12: Descriptive future-period geometric mean loss ratios within the three qualified residential groups, using the reported comparison records. These point summaries do not add group qualifications or simultaneous confidence statements.

Source	Byte volume	Protocol other	Radio quality
Shared stitching	3.16	50.71	43.25
GMM overlay	3.34	48.37	17.42
Latent VAR overlay	3.35	34.80	14.13
Real block	2.08	2.10	1.94

To interpret period changes, compute raw numerical MAE for regression and Brier loss for classification before decomposing each matched task:

$$\Delta \log R = \log(E_{s,2}/E_{s,1}) - \log(E_{r,2}/E_{r,1}).$$

Table S13 averages these within-task log changes using the fixed residential family/group/seed weights. The identity survives this linear averaging. It need not survive the external within-family median. Different physical quantities are not averaged as raw MAE values. The logged changes are dimensionless. Both residential periods refit the generator and predictors, so these are descriptive period contrasts, not a causal drift effect or a fixed-fit transfer test.

Table S13: Descriptive raw-error decomposition for the primary residential task set. Columns are weighted means of within-task log changes from the same to the future period. The final column equals the first minus the second.

Source	Synthetic error	Real-reference error	Relative change
Shared stitching	2.026	1.866	0.160
GMM overlay	1.998	1.866	0.132
Latent VAR overlay	1.844	1.866	-0.022
Real block	2.825	1.866	0.959

For TON_IoT, future direct synthetic skill on the fixed forecasting tasks is 0.299 [0.290, 0.309] for shared stitching, 0.331 [0.299, 0.358] for independent stitching, and 0.468 [0.434, 0.497] for the real contiguous block. The other four reportable sources have negative direct skill. These are recomputed

from each task's synthetic loss and real-training constant-baseline loss, before within-family median and equal-family averaging. They are not obtained by multiplying aggregate ratios and skills. The 5,000-draw seed bootstrap uses seed 814. Failed learned-output jobs remain diagnostic and are excluded.

S8.4 Decision Calculations and Metric Definitions

Residential P0 takes median task log ratios within each family before averaging families. The external implementation instead constructs task-level contrasts before its family median. These aggregation orders are not interchangeable. Section S5 specifies the subsequent quantile calculation.

At least four usable pairs are required separately for calibration and qualification. The reference-skill lower endpoint must be strictly positive. Equality is accepted at the declared order-effect and utility bounds. Numerical checks reproduce all 132 residential decisions and the external-origin decisions under the compared settings.

The metric functions compute per-field KS distance, zero-fraction difference and Wasserstein distance divided by $\max(\text{IQR}, 10^{-6})$. KS uses the recorded SciPy asymptotic method. Temporal profiles average absolute autocorrelation differences at configured row lags, compare unit-sum power spectra from demeaned real FFTs by their L_1 difference, and compare zero/nonzero transition rates. Constant or unsupported comparisons follow the source function's explicit finite/variance checks. Coupling uses the declared pairs, Pearson correlation and nonzero coactivation frequencies. Complete source excerpts preserve the exact scopes, caps, lags and undefined-case handling. The functions do not thereby establish mask-specific or physical validity.

Table S14: Absolute endpoints from the controlled-response curves in the main study. These use the metric implementation specified for those tests, which is separate from the external implementation. The endpoint effect is severity one minus severity zero.

Metric	Changed property	Severity zero	Severity one
KS	Marginal	0	0.315290
Wasserstein/IQR	Marginal	0	1463.912776
Zero-fraction error	Marginal	0	0.037630
C2ST AUC	Marginal	0.505968	1.000000
ACF error	Order	0	0.226208
Spectral L_1	Order	0	1.179905
Transition-rate error	Order	0	0.074330
Correlation error	Coupling	0	0.628144
Coactivation error	Coupling	0	0.014691
C2ST AUC	Coupling	0.505968	0.987874

Table S14 gives the absolute values underlying the normalised curves in main study Figure 2. Section S1 examines the sparse-feature Wasserstein endpoint, and Section S6 lists the response thresholds.

S8.5 Task and Time Coverage

The task register retains the candidate, admission, representative, and membership details described in Section S2. Available class and target-window counts do not enumerate distinct event episodes.

Residential temporal sensitivity excludes input/target windows that cross joins between generated segments for the real-block, independent-stitching, shared-stitching, and latent-VAR sources. This rule concerns 360 recorded temporal comparisons. The primary same-time tasks have zero lag and do not cross these joins. Temporal sensitivity thus describes prediction within segments. It does not evaluate transitions across the joins as if the generated output were one uninterrupted sequence. Gaps in the real acquisition are handled separately from these constructed boundaries.

S8.6 TON_IoT Source Preparation and Checks

The source archive contains 21 filtered CSVs and 21 original logs. The third fridge CSV has 28,200 date entries written as 4 Apr 201 or 04 Apr 201. Its original log gives 4 April 2019 GMT at the corresponding positions. All 153,150 temperature/state pairs match exactly. Using 04 Apr 2019 for these dates, while retaining clock times and measurements, makes every timestamp agree with the original log. The preparation record identifies each affected row and its source timestamp.

The source check covers 2,352,267 filtered records and 2,416,045 original log rows. One malformed original Modbus line is retained as a missing placeholder. Six file pairs have different row counts, and the third garage-door CSV contains 28,934 rows with both measurements blank. These differences mean that the original logs cannot simply replace all filtered files. Preparation retains filtered measurements, missingness, and repeated records. Numeric correspondence is checked at the precision written in each CSV, then observed values are aggregated into UTC-aligned five-second intervals with separate observation counts. Where a tabular timestamp has no explicit zone, UTC alignment is a coordinate convention rather than evidence of physical clock synchronisation.

The packet pipeline processes all 15 normal captures. A separate TShark comparison finds no differences in the tested fields for the first 10,000 complete packets of each file, giving 150,000 checked packets. This sampled comparison supports the decoder checks without claiming exhaustive validation. Within the common analysis period, eight general-network intervals and three IoT-labelled intervals contain no packets. One further IoT-labelled boundary interval is masked because its final record is incomplete. No interval has a reported header-decoding error.

Repeated source downloads have the same reported file sizes and packet counts, and a packet reader identifies an incomplete final record in both files. The preparation rule retains only complete records and excludes the affected interval. It does not infer traffic after the incomplete endings.

Within each capture collection, a gap exceeding 60 seconds starts a new session. Preparation selects the longest session overlapping both capture collections and the telemetry time ranges, retaining complete five-second intervals whose start is included and end excluded. Repeated telemetry records are retained because equal measurements alone do not establish duplicate recording. The resulting table has 28,752 rows and 64 value features. The preparation code records input hashes, selected intervals, exclusion decisions, and output comparisons so that the public analysis can be repeated with the same source files.

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